

Cocrystal Engineering of a Prototype Energetic Material: Supramolecular Chemistry of 2,4,6-Trinitrotoluene

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ABSTRACT: A unique approach to alter the properties of existing energetic materials through cocrystallization is presented. The structure and properties of 17 new cocrystals of the benchmark energetic material 2,4,6-trinitrotoluene (TNT) are elucidated. The donor–acceptor π – π interaction represents a reliable supramolecular synthon for the formation of cocrystals with this aromatic energetic material. When present, amine–nitro interactions direct secondary packing in these crystals. A detailed study of the cocrystals formed reveals an alteration of key properties including density, packing coefficient, melting point, and decomposition temperature. Cocrystallization provides the opportunity to tune properties of existing energetics to create superior materials and provides a complementary alternative to traditional methods that focus primarily on the synthesis of new compounds and/or the discovery of polymorphs with the most attractive properties.

Introduction

The level of effort expended in developing improved energetic materials rivals that involved in drug development. Energetic materials are a class of compounds with high amounts of stored chemical energy, containing both oxidizer and fuel components, that produce high temperature combustion products. This class of materials includes propellants, explosives, and pyrotechnics. In fact, only a very small number of energetic materials have been fielded as high explosives in, for example, military applications because of the need for cost-effective production methods as well as meeting certain rigid performance targets. Therefore, the search for new energetic formulations has often focused on existing chemical entities and either blends and/or polymorphic crystal forms to alter properties such as shock sensitivity, density, melting point, stability, and reactivity. Notably absent from the existing literature is the use of a strategy rapidly gaining ground in pharmaceutical solid form engineering: cocrystallization. Cocrystals are crystals containing two or more neutral molecular components that rely on noncovalent interactions to form a regular arrangement in the solid-state.^{1,2} Through cocrystallization, properties can be dramatically altered, and this technique has therefore been applied extensively to pharmaceuticals to improve solubility, dissolution rate, stability, and bioavailability.^{3–14}

Here cocrystallization is explored as a method to improve properties critical to energetic materials. Through the formation of cocrystals, commonly used energetic materials might be enhanced and existing energetic materials that currently fail to meet all of the requirements for application could be made viable. For example, the energetic materials nitroglycerin and triacetone triperoxide (TATP) possess desirable qualities such as high power and inexpensive synthesis, respectively, but due to their high shock sensitivity, neither are used in their pure state. Through cocrystallization, such materials could be stabilized, allowing for their common use.

Much of the existing literature on cocrystal formation focuses on the exploitation of classical hydrogen bonding schemes.^{15,16}

The requisite functional groups for such interactions are generally lacking in energetic materials, thus necessitating the exploration of alternative supramolecular synthons. Some multicomponent complexes of energetic materials are reported in the literature, though very few crystal structures are available and the data on the interactions driving the formation of these complexes is limited. Although a considerable number of solvates of both cyclotetramethylenetetranitramine (HMX) and cyclotrimethylenetrinitramine (RDX) were discovered during efforts to purify HMX from RDX, only five solvated structures are reported for HMX and one for RDX.^{17–20} Multicomponent complexes, both in solution and in the solid state, of aromatic energetic materials including TNT and hexanitrostilbene (HNS), are also reported in the literature.^{21–29} From this work there is only one crystal structure of a TNT cocrystal reported in the Cambridge Structural Database (CSD), TNT/pyrene.³⁰ This cocrystal features donor–acceptor π – π interactions, leading to the 1:1 complex. Here the generality of TNT cocrystallization is scrutinized for a wide range of cocrystal formers (ccf), and the alteration of structural, chemical, and spectroscopic properties, especially as they are relevant to energetic materials, is elucidated.

Results and Discussion

Structure of TNT Cocrystals. The following cocrystals formed with TNT: naphthalene (1), 1-bromonaphthalene (2), anthracene (3), 9-bromoanthracene (4), phenanthrene (5), perylene (6), tetrathiafulvalene (7), thieno[3,2-*b*]thiophene (8), phenothiazine (9), dibenzothiophene (10), 4,6-dimethyldibenzothiophene (11), 1,2-phenylenediamine (12), 1,4-dimethoxybenzene (13), 4-aminobenzoic acid (14, 15), and anthranilic acid (16, 17). All of the cocrystals contain a 1:1 ratio of components, with the exception of 4-aminobenzoic acid and anthranilic acid, which exist in both a 1:1 and 1:2 ratio with TNT (TNT/ccf). The crystallographic data of the TNT cocrystals obtained are presented in Table 1.

Donor–acceptor π – π interactions drive the formation of these cocrystals. Due to the strong polarizing effects of the nitro

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Table 1. Crystallographic Information for the 17 Cocrystals of TNT^a

	1	2	3	4	5	6	7	8	9
formula	C ₁₇ H ₁₃ N ₃ O ₆	C ₁₇ H ₁₂ BrN ₃ O ₆	C ₂₁ H ₁₅ N ₃ O ₆	C ₂₁ H ₁₄ BrN ₃ O ₆	C ₂₁ H ₁₅ N ₃ O ₆	C ₂₇ H ₁₇ N ₃ O ₆	C ₁₃ H ₉ N ₃ O ₆ S ₄	C ₁₃ H ₉ N ₃ O ₆ S ₂	C ₁₉ H ₁₄ N ₄ O ₆ S
MW (g/mol)	355.30	434.21	405.36	484.26	405.36	479.44	431.47	367.35	426.40
stoichiometry	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1	1:1
color	colorless	light yellow	orange	yellow	light yellow	red	dark brown	light yellow	dark brown
morphology	blade	blade	blade	blade	blade	needle	prism	blade	blade
solvent	ethanol	neat	THF	THF	ethanol	toluene	THF	ethanol	ethanol
crystal system	triclinic	triclinic	monoclinic	triclinic	orthorhombic	orthorhombic	triclinic	triclinic	orthorhombic
space group	<i>P</i> $\bar{1}$	<i>P</i> $\bar{1}$	<i>P</i> ₂₁ / <i>c</i>	<i>P</i> $\bar{1}$	<i>P</i> ₂₁ ₂₁ ₂₁	<i>P</i> ₂₁ ₂₁ ₂₁	<i>P</i> $\bar{1}$	<i>P</i> $\bar{1}$	<i>P</i> ₂₁ ₂₁ ₂₁
<i>a</i> (Å)	8.6559(6)	7.2538(3)	14.0050(3)	6.7461(2)	7.07587(16)	7.19649(17)	8.39611(15)	6.81044(18)	7.4869(4)
<i>b</i> (Å)	9.8012(5)	7.7969(3)	6.72457(16)	8.61691(19)	9.2294(2)	7.98291(18)	9.42210(19)	9.2640(3)	9.1121(4)
<i>c</i> (Å)	10.0969(6)	14.8572(5)	19.4127(5)	17.3981(4)	27.7299(7)	36.206(3)	11.4671(2)	12.7021(4)	26.5398(19)
α (deg)	92.333(3)	91.504(3)	90.00	97.606(2)	90.00	90.00	86.4170(10)	95.909(2)	90.00
β (deg)	111.371(4)	92.976(2)	103.538(2)	100.297(2)	90.00	90.00	69.5530(10)	91.239(2)	90.00
γ (deg)	96.598(4)	98.164(3)	90.00	103.182(2)	90.00	90.00	81.0340(10)	113.853(2)	90.00
cell vol (Å ³)	789.26(8)	830.17(6)	1777.44(7)	952.90(4)	1810.92(8)	2079.98(16)	839.58(3)	727.29(4)	1810.56(18)
ρ_{calc} (g/cm ³)	1.495	1.737	1.515	1.698	1.487	1.531	1.707	1.677	1.564
<i>Z</i>	2	2	4	2	4	4	2	2	4
data/param	2479/236	2581/245	3118/272	3075/280	3113/308	3078/296	2682/236	2302/218	2535/307
<i>R</i> ₁ / <i>wR</i> ₂	3.8/11.0	5.8/17.2	4.6/11.3	3.8/9.6	8.8/22.7	8.5/24.8	2.7/6.9	3.7/12.0	8.9/18.8
GOF	1.109	1.235	1.122	1.073	1.096	1.132	1.056	1.189	1.223
	10	11	12	13	14	15	16	17	
formula	C ₁₉ H ₁₃ N ₃ O ₆ S	C ₂₁ H ₁₇ N ₃ O ₆ S	C ₁₃ H ₁₃ N ₅ O ₆	C ₁₅ H ₁₅ N ₃ O ₈	C ₁₄ H ₁₂ N ₄ O ₈	C ₂₁ H ₁₉ N ₅ O ₁₀	C ₁₄ H ₁₂ N ₄ O ₈	C ₂₁ H ₁₉ N ₅ O ₁₀	
MW (g/mol)	411.38	439.44	335.28	365.30	364.28	501.41	364.28	501.41	
stoichiometry	1:1	1:1	1:1	1:1	1:1	1:2	1:1	1:2	
color	light yellow	yellow	brown	yellow	yellow	orange	yellow	orange	
morphology	blade	blade	blade	blade	blade	prism	blade	prism	
solvent	ethanol	ethanol	ethanol	ethanol	ethanol	ethanol	ethanol	melt	
crystal system	monoclinic	triclinic	orthorhombic	monoclinic	monoclinic	monoclinic	triclinic	triclinic	
space group	<i>P</i> ₂₁ / <i>c</i>	<i>P</i> $\bar{1}$	<i>Pna</i> 2 ₁	<i>P</i> ₂₁ / <i>n</i>	<i>P</i> ₂₁	<i>P</i> ₂₁ / <i>c</i>	<i>P</i> $\bar{1}$	<i>P</i> $\bar{1}$	
<i>a</i> (Å)	9.09248(16)	7.22567(18)	19.0408(13)	6.6019(8)	6.90525(18)	10.5492(4)	6.68253(12)	10.3188(3)	
<i>b</i> (Å)	7.17969(18)	9.3056(2)	6.66738(12)	28.237(3)	14.1124(3)	14.4420(5)	8.67582(16)	13.8350(3)	
<i>c</i> (Å)	54.4492(10)	15.1173(3)	11.1188(2)	8.6743(8)	15.7976(11)	14.7554(6)	14.2195(10)	15.3114(11)	
α (deg)	90.00	104.5420(10)	90.00	90.00	90.00	90.00	86.507(6)	89.339(6)	
β (deg)	95.7200(10)	96.8790(10)	90.00	91.197(7)	93.559(7)	101.061(2)	80.859(6)	85.674(6)	
γ (deg)	90.00	91.3760(10)	90.00	90.00	90.00	90.00	68.812(5)	81.667(6)	
cell vol (Å ³)	3536.81(13)	975.28(4)	1411.56(11)	1616.7(3)	1536.50(12)	2206.25(15)	758.90(6)	2156.63(17)	
ρ_{calc} (g/cm ³)	1.545	1.496	1.578	1.501	1.575	1.510	1.594	1.544	
<i>Z</i>	8	2	4	4	4	4	2	4	
data/param	6058/524	3145/281	2366/233	2511/235	4686/494	3764/351	2557/235	7458/697	
<i>R</i> ₁ / <i>wR</i> ₂	5.9/15.2	5.9/15.8	3.4/10.0	6.5/16.9	5.8/12.8	7.5/17.7	5.1/14.3	9.4/25.3	
GOF	1.056	1.159	0.981	1.032	1.113	1.154	1.002	1.042	

^a All structures were collected at 95 K.

groups, TNT has an electron-poor π -system; conversely, the cocrystal formers all possess relatively electron-rich benzene rings, promoting a favorable interaction between the electron-poor and electron-rich aromatic rings. The calculated electrostatic potential surfaces of TNT and the cocrystal formers are shown in Figure 1.

The introduction of donor–acceptor π – π interactions between aromatic rings has a profound influence on the packing of the molecules in these cocrystals and, consequently, on the solid-state properties. Both polymorphs of TNT, as well as most of the cocrystal formers in their pure form, pack in a herringbone motif to facilitate electrostatically favorable edge-to-face interactions.^{31–33} Due to the donor–acceptor π – π interactions, face-to-face π -stacking is observed in the cocrystals. Representative examples of both packing motifs are presented in Figure 2. All of the TNT cocrystals exhibited similar face-to-face π -stacking motifs to the TNT/anthracene cocrystal (see Supporting Information).

The formation of cocrystals with TNT not only alters the π -stacking motif of the cocrystal formers but also significantly reorients the nitro groups of the TNT molecule. In

particular, there is a notable and consistent change in the angle of the nitro group in the 4 position (para to the CH₃) of TNT in the cocrystals. In monoclinic TNT these nitro groups of the inequivalent molecules make angles of 33.8 and 23.7° to the benzene ring, whereas in orthorhombic TNT the corresponding angles are 33.9 and 23.2°. In the TNT cocrystals, these angles range from 1.3 to 14.3° and are 5.0° on average. Torsion angles are presented in Table 2.

The change in orientation of the nitro groups influences the properties of TNT within the cocrystals. Indeed, this change is significant enough that it results in shifts in the vibrational spectra of these materials. Monoclinic and orthorhombic TNT have an asymmetric nitro stretch (ν_{as}) at 1537.4 and 1535.7 cm^{−1} and a symmetric nitro stretch (ν_{s}) at 1368.7 and 1358.7 cm^{−1}, respectively. In the TNT cocrystals, the asymmetric nitro stretch generally increases in energy with respect to both forms of TNT, 6.7 and 8.4 cm^{−1} on average for monoclinic and orthorhombic TNT, respectively. In contrast to this, the symmetric nitro stretch in the TNT cocrystals decreases in energy, 8.7 cm^{−1} on average, with respect to monoclinic TNT, while remaining relatively unchanged with respect to orthorhombic TNT.³⁴

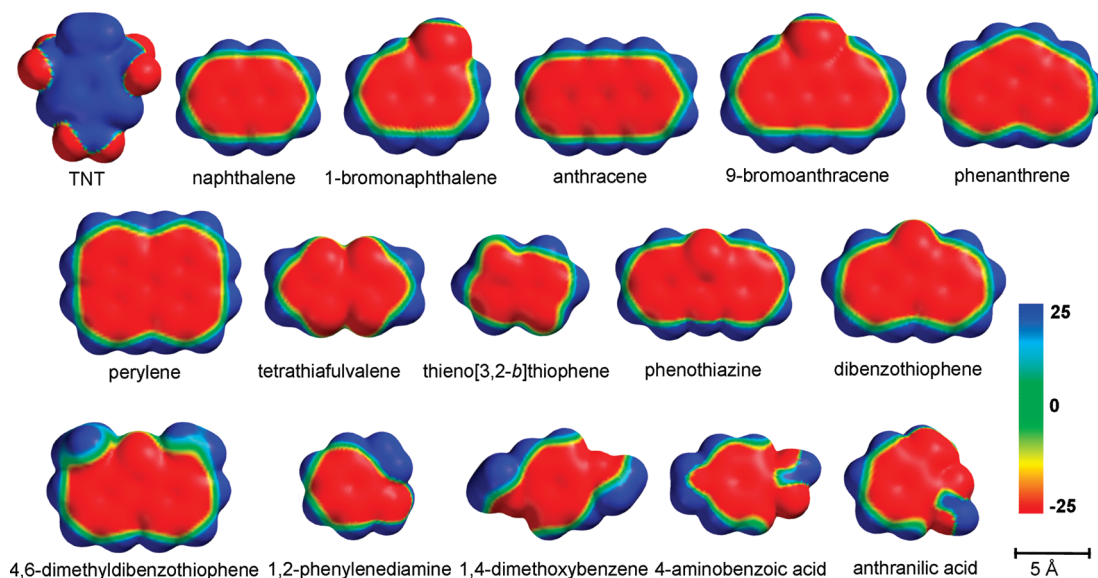


Figure 1. Electrostatic potential surfaces of TNT and the cocystal formers calculated using the semiempirical method and the AM1 model. Red and blue surfaces represent electron rich and poor regions, respectively. Surfaces are normalized between -25 and 25 kJ/mol.

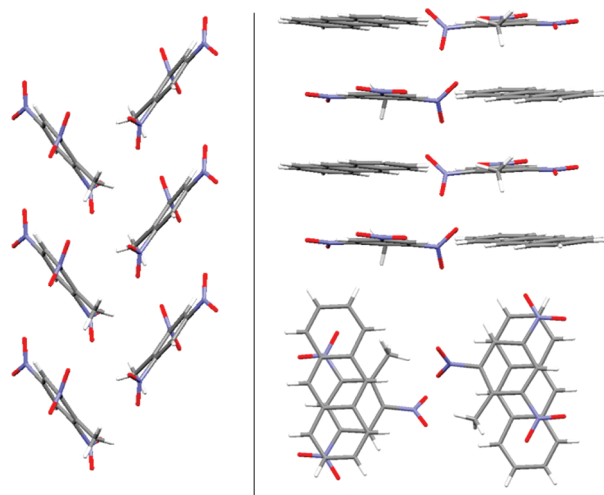


Figure 2. Comparison of the herringbone packing of monoclinic TNT (CSD code ZZZMUC08)³³ (left) and the face-to-face π -stacking of the TNT/anthracene cocystal (right).

Although the donor–acceptor π – π interactions are dominant in the formation of the TNT cocystals reported herein, the secondary interactions within these cocystals provide valuable insight into favorable interactions that can be applied in the formation of cocystals with other nitrated energetic materials. The amine–nitro interaction is expected to be among the strongest of the interactions found in the TNT cocystals and indeed is present in all of the cocystals containing amine functional groups, including the cocystals of TNT with 1,2-phenylenediamine, 4-aminobenzoic acid, anthranilic acid, and phenothiazine.³⁵ Monocoordinated amine–nitro interactions are observed in almost every TNT cocystal containing an amine functional group (examples shown in Figure 3). These interactions are incorporated into a wide variety of motifs in the TNT cocystals, including dimers, chains, and rings. Bicoordinated amine–nitro interactions occur only in the TNT/anthranilic acid 1:2, TNT/1,2-phenylenediamine, and TNT/phenothiazine cocystals (Figures 4 and 5).

Table 2. Average Torsion Angles for the 4-Nitro Group of TNT in the TNT Polymorphs and Cocystals^a

TNT/TNT cocystal	TNT A (deg)	TNT B (deg)
monoclinic TNT	33.8	23.7
orthorhombic TNT	33.9	23.2
naphthalene	5.4	
1-bromonaphthalene	2.1	
anthracene	3.4	
9-bromoanthracene	2.3	
phenanthrene	1.3	
perylene	2.5	
tetrathiafulvalene	3.5	
thieno[3,2- <i>b</i>]thiophene	4.8	
phenothiazine	4.1	
dibenzothiophene	1.4	2.3
4,6-dimethyldibenzothiophene	4.5	
1,2-phenylenediamine	1.8	
1,4-dimethoxybenzene	4.7	
4-aminobenzoic acid 1:1	9.0	14.3
4-aminobenzoic acid 1:2	6.4	
anthranilic acid 1:1	6.2	
anthranilic acid 1:2	4.7	14.6

^aWhen two inequivalent TNT molecules are present, both torsion angles are given.

The monocoordinated amine–nitro interactions in the TNT/1,2-phenylenediamine cocystal form a chain ($C_2^2(10)$) that generates sheets of alternating TNT and 1,2-phenylenediamine molecules when linked through the bicoordinated (2.376 and 2.450 Å) amine–nitro interaction ($R_1^2(4)$). In the TNT/anthranilic 1:2 cocystal, the bicoordinated amine–nitro interaction (2.221 and 2.397 Å) forms a dimer between anthranilic acid and TNT ($R_1^2(4)$). In a similar fashion, the amine of the TNT/phenothiazine cocystal is bicoordinated to a nitro group of TNT, though asymmetrically (2.381 and 2.754 Å), to form a dimer ($R_1^2(4)$).

In general, the TNT cocystals containing amines do not preferentially form bicoordinated (three-centered) amine–nitro interactions, as might be expected based on interactions of nitroanilines reported in the literature.^{16,36}

Properties of TNT Cocystals. One significant result of cocrystallizing TNT is that a relatively dramatic change in the density of this material can be accomplished. Density is

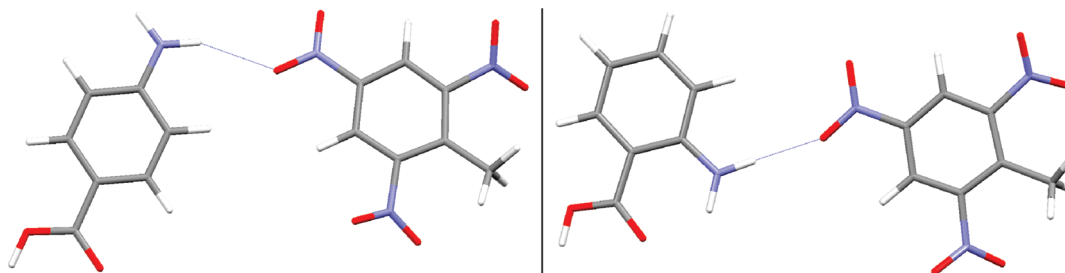


Figure 3. Example of a monocoordinated amine–nitro interaction in the TNT/4-aminobenzoic acid 1:1 cocrystal (left) and the TNT/anthranilic acid 1:1 cocrystal (right).

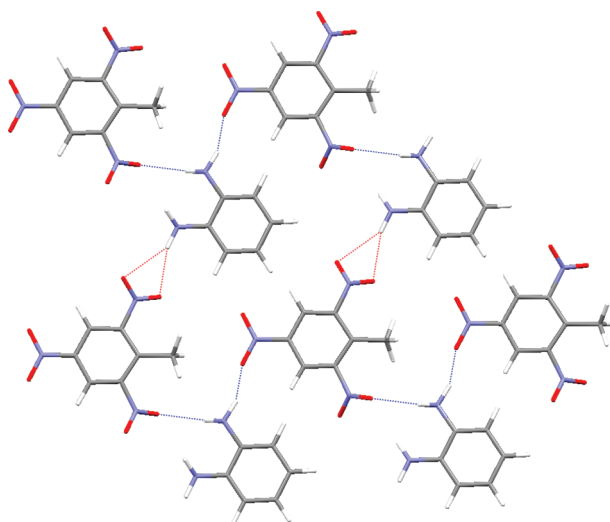


Figure 4. Amine–nitro interactions in the TNT/1,2-phenylenediamine cocrystal. Both monocoordinated and bicoordinated amine–nitro interactions are observed.

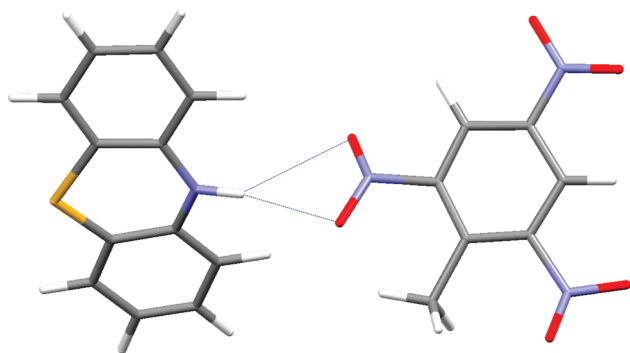


Figure 5. Bicoordinated amine–nitro interaction in the major conformer of the TNT/phenothiazine cocrystal.

often used as a very general guide to predict properties, including the detonation velocity and detonation pressure, of explosives.³⁷ More dense energetic materials have the potential to have larger detonation velocities and therefore are preferred. A comparison of the densities of the TNT cocrystals to that of the cocrystal formers is presented in Table 3.

As might be expected, many of these cocrystals have a density in between those of both components. This is not always the case, however; the TNT/tetrathiafulvalene cocrystal has a density higher than those of two of the reported forms of tetrathiafulvalene and the orthorhombic form of TNT. Perhaps more notable is the TNT/1-bromonaphthalene cocrystal, in which the density is higher than those of both forms of TNT as well as pure

Table 3. Comparison of the Densities of the TNT Cocrystals to the Available Polymorphs of the Cocrystal Formers (ccf)^{38–57a}

compound	TNT cocrystal (g/cm ³)	ccf form I (g/cm ³)	ccf form II (g/cm ³)	ccf form III (g/cm ³)
phenanthrene	1.487	1.222 ^d		
naphthalene	1.495	1.239 ^m		
4,6-dimethyldibenzo- thiophene	1.496	1.299 ^d		
1,4-dimethoxybenzene	1.501	1.231 ^g		
4-aminobenzoic acid 1:2	1.509	1.373 ^d	1.389 ^{c, d}	
anthracene	1.515	1.297 ^f	1.250 ^d	
perylene	1.531	1.388 ^g	1.394 ^f	
anthranilic acid 1:2	1.544	1.369 ^d	1.409 ^d	1.388 ^d
dibenzothiophene	1.545	1.418 ^f		
phenothiazine	1.561	1.387 ^d	1.417 ⁱ	
4-aminobenzoic acid 1:1	1.575	1.373 ^d	1.389 ^{c, d}	
1,2-phenylenediamine	1.578	1.214 ^d		
anthranilic acid 1:1	1.594	1.369 ^d	1.409 ^d	1.388 ^d
thieno[3,2- <i>b</i>]thiophene	1.677	1.542 ^d		
9-bromoanthracene	1.698	1.592 ^d	1.622 ^d	
TNT		1.713 ^j	1.704 ^h	
tetrathiafulvalene	1.707	1.686 ^e	1.750 ^k	1.676 ^j
1-bromonaphthalene	1.737	1.489 ^b		

^a All densities reported are for crystals at 95 K unless otherwise specified. ^b Liquid at room temperature; no crystal structure available. ^c Designated form beta IV. ^d Room temperature. ^e 290 K. ^f 200 K. ^g 150 K. ^h 123 K. ⁱ 120 K. ^j 100 K. ^k 98 K. ^l 94 K. ^m 90 K.

1-bromonaphthalene. This latter cocrystal, in particular, demonstrates the effectiveness of cocrystallization in achieving higher density materials. It should be noted, however, when cocrystallizing an energetic material with a nonenergetic material that the relationship of density to the detonation properties no longer applies. In the case of the TNT/1-bromonaphthalene cocrystal, even though the overall density of 1.737 g/cm³ is greater than that of monoclinic TNT (1.713 g/cm³), the effective density of TNT is reduced to 0.909 g/cm³. While TNT is diluted in this case, these results still raise the intriguing possibility that, by cocrystallizing two energetic materials together, higher density explosives with enhanced performance could be achieved.

In the case of TNT/1-bromonaphthalene, not only has the density increased, but multifunctionality has been added to this material as well. Through cocrystallization, a halogen, bromine, is effectively added to TNT. The production of reactive halogen species within the explosion plume provides a means of annihilating biological and chemical warfare agents (BWA and CWA) remotely.^{58,59} Much effort has been exerted to synthetically introduce halogens into energetics to create effective anti-BWA and anti-CWA. Here cocrystallization is shown to provide a straightforward means to add

Table 4. Comparison of the Packing Coefficients of the TNT Cocrystals to the Available Polymorphs of the Cocrystal Formers (ccf)^{38–57a}

compound	TNT cocrystal	ccf form I	ccf form II	ccf form III
1,4-dimethoxybenzene	81.6%	82.2% ^g		
anthranilic acid 1:2	82.0%	81.2% ^d	83.8% ^d	82.3% ^d
4-aminobenzoic acid 1:1	82.2%	82.1% ^d	83.2% ^{c,d}	
4-aminobenzoic acid 1:2	82.2%	82.1% ^d	83.2% ^{c,d}	
TNT		82.4% ^j	81.9% ^h	
anthranilic acid 1:1	82.6%	81.2% ^d	83.8% ^d	82.3% ^d
phenanthrene	82.7%	82.1% ^d		
4,6-dimethyldibenzothiophene	82.7%	83.1% ^d		
phenothiazine	82.8%	84.3% ^d	86.1% ⁱ	
9-bromoanthracene	83.1%	81.8% ^d	86.1% ^d	
dibenzothiophene	83.2%	87.4% ^j		
thieno[3,2- <i>b</i>]thiophene	83.4%	82.9% ^d		
1-bromonaphthalene ^b	83.5%			
naphthalene	84.0%	87.5% ^m		
anthracene	85.0%	88.1% ^l	85.0% ^d	
1,2-phenylenediamine	85.4%	80.7% ^d		
tetrathiafulvalene	85.4%	80.4% ^e	83.8% ^k	79.9% ^j
perylene	86.6%	89.6% ^g	90.0% ^f	

^a All packing coefficients reported are for crystal forms at 95 K unless otherwise specified. ^b Liquid at room temperature; no crystal structure available. ^c Designated form beta IV. ^d Room temperature. ^e 290 K. ^f 200 K. ^g 150 K. ^h 123 K. ⁱ 120 K. ^j 100 K. ^k 98 K. ^l 94 K. ^m 90 K.

halogens to existing energetic materials. By selecting appropriately halogenated cocrystal formers, cocrystallization offers the opportunity to modify energetics as needed.

In addition to density, cocrystallization also changes the packing coefficient. As evidenced from the packing coefficients of the TNT cocrystals and cocrystal formers in Table 4, all of the packing coefficients are altered with respect to the cocrystal formers. Calculation of packing coefficients is described in the Supporting Information.

While many of the TNT cocrystals do not possess higher density with respect to TNT, the packing coefficient of TNT improves remarkably, by as much as 3.6%, through cocrystallization. Indeed, the packing coefficients of more than half of the cocrystals formed are higher than those of either polymorph of TNT, indicating a strong complementary relationship between the molecules. Although there is a reduction of the packing coefficient of the cocrystals with respect to the solid forms of many of the aromatic compounds without functional groups, these compounds provide a means to improve the packing coefficient with respect to the solid forms of TNT by minimizing the empty space within the crystal lattice of this energetic material.^{60,61} These cocrystals highlight the potential to minimize the free volume in solid forms with TNT and other energetic materials and provide insights into the size and shapes of molecules that can improve the packing coefficient.

Cocrystallization forms unique compounds, and therefore, drastically altered melting points can be achieved. A summary of the melting points for the TNT cocrystals is presented in Table 5. The TNT cocrystals formed exhibit a wide range of melting points. Through cocrystallization, the melting point with respect to TNT can be lowered by as much as 36 °C or increased by 94 °C. While many of the melting points of the cocrystals formed are between that of both cocrystal formers, this is not always the case. Two notable exceptions to this trend are the TNT cocrystals with 1,4-dimethoxybenzene and 1,2-phenylenediamine, where the cocrystal exhibits a lower melting point than those of either of the pure components. By contrast, the cocrystals of TNT

Table 5. Melting Points of TNT Cocrystals as Compared to the Most Stable Polymorph of the Cocrystal Formers (ccf) Measured Using Differential Scanning Calorimetry (DSC) at a Ramp Rate of 10 °C/min

TNT/TNT cocrystal	ccf (°C)	cocrystal (°C)
1,4-dimethoxybenzene	56.6	45.2
1-bromonaphthalene	−1.0	73.3
1,2-phenylenediamine	101.8	73.8
orthorhombic TNT	81.1	
monoclinic TNT	81.9	
thieno[3,2- <i>b</i>]thiophene	56.0	83.7
naphthalene	81.5	95.9
phenanthrene	99.7	100.2
9-bromoanthracene	102.9	101.0
phenothiazine	186.4	107.2
anthracene	217.1	108.5
tetrathiafulvalene	119.2	108.8
4,6-dimethyldibenzothiophene	154.6	116.3
4-aminobenzoic acid 1:1	188.1	117.6
dibenzothiophene	99.0	118.8
anthranilic acid 1:2	99.0	139.3
perylene	276.0	151.5
anthranilic acid 1:1	99.0	151.7
4-aminobenzoic acid 1:2	188.1	175.2

Table 6. Decomposition Temperature and Enthalpy of TNT and TNT Cocrystals as Measured by DSC at a Ramp Rate of 10 °C/min

TNT/TNT cocrystal	<i>T</i> _{decomp} (°C)	−Δ <i>H</i> _{decomp} (kJ/g)
tetrathiafulvalene	202.9	0.95
phenothiazine	237.3	1.1
4-aminobenzoic acid 1:2	251.8	0.34
4-aminobenzoic acid 1:1	256.7	1.0
anthranilic acid 1:2	258.4	0.45
anthranilic acid 1:1	266.2	1.0
1,2-phenylenediamine	267.3	0.91
perylene	268.6	0.91
9-bromoanthracene	288.2	1.5
1,4-dimethoxybenzene	289.5	0.89
thieno[3,2- <i>b</i>]thiophene	291.9	1.3
anthracene	293.5	1.4
1-bromonaphthalene	296.4	0.84
monoclinic TNT	298.0	1.7
4,6-dimethyldibenzothiophene	299.1	1.4
orthorhombic TNT	299.6	2.0
naphthalene	304.2	0.71
phenanthrene	307.0	1.3
dibenzothiophene	312.8	1.2

with thieno[3,2-*b*]thiophene, naphthalene, dibenzothiophene, and anthranilic acid all exhibit melting points higher than those of either cocrystal former. Through cocrystallization, therefore, it is possible to manipulate the melting point of energetic materials as needed. For example, decreasing the melting point can be desirable to form energetic materials that can be easily melt-processed. Conversely, increasing the melting point can be advantageous when creating munitions that resist deformation prior to detonation. The range of temperatures available through this technique is notably broader than that achievable through polymorphic modification.

Cocrystallization also can produce significant changes in another critical thermal property for energetic materials, namely the decomposition temperature. Ideally this value is high for energetic materials, to ease processing as well as to increase their detonation temperature and thus allow for deeper penetration of the energetic material into a target. The decomposition temperature data are summarized in Table 6. In almost every case, the decomposition temperature of cocrystals was altered with respect to the case of TNT. Even in the liquid state, the interactions driving the formation of the cocrystals are significant enough to change the decomposition temperature

by as much as 95 °C. It is notable that, in some of the cocrystals formed, the decomposition temperature can be increased with respect to that of TNT. This is observed in TNT cocrystals with phenanthrene, naphthalene, and dibenzothiophene. By selecting the appropriate cocrystal former, therefore, cocrystallization provides a means of increasing the decomposition temperature and therefore improving this aspect of a common energetic material.

Although the TNT cocrystals presented here are diluted through cocrystallization with nonenergetic materials, violent decomposition is still achievable. In all DSC thermograms, a substantial heat of decomposition was observed. Qualitative tests of a subset of cocrystals confirm that violent decomposition is still achievable in these TNT cocrystals with nonenergetics, thus validating the approach of making new multifunctional energetic compositions based on a single energetic component.

Conclusions

A wide range of aromatic compounds successfully form cocrystals with TNT mediated by donor–acceptor π – π interactions. These interactions direct the cocrystal structure, giving preference to face-to-face over edge-to-face stacking. When present, amine–nitro interactions also guide the cocrystal structure. The alterations in structure ultimately produce unique properties in the cocrystals, with respect to the pure components. Properties vital to energetic materials, including the density, packing coefficient, melting point, and decomposition temperature are successfully modified through cocrystallization. Indeed, this technique allows for wider variations in the properties of energetic materials than is achievable through polymorphic modification. Although the interactions mediating cocrystal formation discussed here are applicable primarily to aromatics, the concept of cocrystallizing energetic materials to improve their properties can be applied broadly.

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Supporting Information Available: Experimental methods, Raman spectra of TNT cocrystals, nitro stretching frequencies in Raman spectra, crystallographic data of TNT cocrystals, and X-ray crystallographic information files (CIF) for compounds 1–17. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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